

D665

Learner Development and the Science of Learning

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A nine-year-old who wears glasses and has a younger sister often gives up on projects, saying her work is never as good as others'. According to Erikson, which developmental task is she working through?

- A. Identity vs. role confusion
- B. Sibling rivalry stage
- C. Trust vs. mistrust
- D. Industry vs. inferiority

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A teacher of 15-year-olds wants to support Erikson's main developmental task for that age. Which activity fits best?

- A. Activities focused on learning to trust caregivers.
- B. Projects where students explore values, interests, and possible future roles.
- C. Any activity, since all stages end by age 12.
- D. Sticker charts for finished worksheets.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A bright 14-year-old on the soccer team knows the long-term project is due in three weeks but does nothing until the night before, then feels overwhelmed. Her parents both work as nurses. What is the best teacher response?

- A. Ask her parents to manage it because they are health workers.
- B. Treat it as laziness and lower her grade.
- C. Assume she will plan well by herself at this age.
- D. Break the project into checkpoints with due dates and teach a planning routine.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A high school student plans to stay up all night studying for a biology test. He drinks two sodas a day and takes the bus at 6:45 a.m. What does learning science suggest?

- A. Study earlier and get a full night's sleep, since sleep helps lock in what was learned.
- B. Stay up, because more hours of study always mean more learning.
- C. Cut out the sodas and stay up.
- D. His ability is fixed, so study time does not matter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A seventh grader says, "I'm just not a math person." He enjoys skateboarding and helps at his family's store. According to Bandura, which approach will most strongly build his self-efficacy in math?

- A. Give him carefully sequenced problems he can succeed at, building to harder ones.
- B. Tell him often that he is smart.
- C. Have him skip math and focus on skateboarding.
- D. Show him a video of a math professor solving hard proofs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A teacher and a parent hold monthly meetings to align reading routines at home and school. The parent works nights at a warehouse, and the child has a pet rabbit. In Bronfenbrenner's model, the teacher-parent connection is part of which system?

- A. Exosystem
- B. Macrosystem
- C. Mesosystem
- D. Microsystem

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A child is not directly part of a setting, but decisions made there affect her—for example, her parent's employer changes the parent's work hours. Which system is this?

- A.** Microsystem
- B.** Mesosystem
- C.** Exosystem
- D.** Chronosystem

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher wants students to keep reading after the class contest ends. The classroom library has 400 books, and the contest prize was a pizza party. Which approach best supports lasting motivation?

- A.** Hold another pizza party contest every month.
- B.** Rewards always increase lasting motivation.
- C.** Let students choose books, set personal reading goals, and share favorites in book clubs.
- D.** Assign every book so students do not waste time choosing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Many fifth graders believe heavier objects fall faster. The class has a new set of balances, and the teacher's lesson is 45 minutes long. Which approach best reflects conceptual change teaching?

- A.** Tell students the right answer and move on.
- B.** Have students predict, then drop objects of different mass together, observe, and discuss why the prediction did not hold.
- C.** Ignore the misconception so students are not confused.
- D.** Use the balances to weigh each object.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

After a unit, students can repeat the textbook definition of photosynthesis but still say plants get their food from the soil. What does this suggest?

- A.** Misconceptions cannot ever be changed.
- B.** The misconception was not surfaced and challenged, so it persists alongside the memorized definition.
- C.** Students fully understand photosynthesis.
- D.** They need to memorize the definition more.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher is planning a 5E lesson on circuits. The school just bought battery kits, and the class meets after recess. After an Engage hook showing a flashlight that will not light, what should come next?

- A.** Students explore with batteries, bulbs, and wires to try to light the bulb.
- B.** Students apply circuits to design an alarm.
- C.** A quiz on circuit vocabulary.
- D.** The teacher explains how circuits work with a diagram.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

In a 5E lesson, students use what they learned about circuits to design a doorbell for a model house. Which phase is this?

- A.** Explain
- B.** Explore
- C.** Engage
- D.** Elaborate

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A nine-year-old ~~who wears glasses and has a younger sister~~ often gives up on projects, saying her work is never as good as others'. According to Erikson, which developmental task is she working through?

Target word: Erikson stage **Decides:** Age 9 and feelings of competence.

A	Adjacent category	That is the adolescent stage.
B	Camouflage	Not an Erikson stage; reacts to the sister detail.
C	Confusion set	An infancy stage.
D	Correct	School-age children build competence or feel inferior.

2. Answer B Concept 1 · transfer

A teacher of 15-year-olds wants to support Erikson's main developmental task for that age. Which activity fits best?

Target word: reverse: activity from stage **Decides:** Adolescents work on identity.

A	Confusion set	Infancy task.
B	Correct	Supports identity formation.
C	Absolute language	Development continues through adolescence.
D	Inverted operation	Targets younger children's industry needs.

3. Answer D Concept 2

A bright 14-year-old ~~on the soccer team~~ knows the long-term project is due in three weeks but does nothing until the night before, then feels overwhelmed. ~~Her parents both work as nurses.~~ What is the best teacher response?

Target word: support for developing executive function **Decides:** Planning skills are still developing in adolescence.

A	Camouflage	Their jobs do not decide the classroom response.
B	Adjacent category	Frames a developmental skill gap as character.
C	Absolute language	Executive function matures into the mid-20s.
D	Correct	External structure supports a still-developing prefrontal cortex.

4. Answer A Concept 3

A high school student plans to stay up all night studying for a biology test. ~~He drinks two sodas a day and takes the bus at 6:45 a.m.~~ What does learning science suggest?

Target word: sleep and learning **Decides:** Sleep consolidates memory.

A	Correct	Sleep supports memory consolidation.
B	Absolute language	Sleep loss impairs memory and attention.
C	Camouflage	Addresses a side detail but keeps the sleep loss.
D	Confusion set	Neuroplasticity means practice changes the brain.

5. Answer A Concept 4

A seventh grader says, "I'm just not a math person." ~~He enjoys skateboarding and helps at his family's store.~~ According to Bandura, which approach will most strongly build his self-efficacy in math?

Target word: strongest source of self-efficacy **Decides:** Mastery experiences are the strongest source.

A	Correct	Mastery experiences are the strongest source.
B	Adjacent category	Verbal persuasion is a weaker source.
C	Camouflage	Avoids the skill instead of building efficacy.
D	Dropped element	Modeling works best when the model is similar to the learner.

6. Answer C Concept 5

A teacher and a parent hold monthly meetings to align reading routines at home and school. ~~The parent works nights at a warehouse, and the child has a pet rabbit.~~ In Bronfenbrenner's model, the teacher-parent connection is part of which system?

Target word: classify the system **Decides:** A link between two microsystems.

A	Camouflage	The parent's job is exosystem, but that is not what is asked.
B	Adjacent category	Cultural values and broad systems.
C	Correct	Connections between the child's direct settings.
D	Confusion set	Microsystems are the settings themselves, not the links.

7. Answer C Concept 5 · transfer

A child is not directly part of a setting, but decisions made there affect her—for example, her parent's employer changes the parent's work hours. Which system is this?

Target word: reverse: from description to system **Decides:** Indirect influence.

A	Inverted operation	Direct settings the child participates in.
B	Confusion set	Links between the child's direct settings.
C	Correct	Settings the child is not in that still affect her.
D	Adjacent category	Changes over time.

8. Answer C Concept 6

A teacher wants students to keep reading after the class contest ends. ~~The classroom library has 400 books, and the contest prize was a pizza party.~~ Which approach best supports lasting motivation?

Target word: self-determination theory **Decides:** Autonomy, competence, relatedness sustain motivation.

A	Camouflage	External rewards alone can weaken intrinsic motivation.
B	Absolute language	Rewards can undercut interest.
C	Correct	Builds autonomy, competence, and relatedness.
D	Inverted operation	Removes autonomy.

9. Answer B Concept 7

Many fifth graders believe heavier objects fall faster. ~~The class has a new set of balances, and the teacher's lesson is 45 minutes long.~~ Which approach best reflects conceptual change teaching?

Target word: addressing a misconception **Decides:** Surface the idea and confront it with evidence.

A	Dropped element	Telling skips the step that changes the prior idea.
B	Correct	Surfacing the idea and creating conflicting evidence drives change.
C	Inverted operation	Unaddressed ideas persist.
D	Camouflage	Weighing does not test fall speed.

10. Answer B Concept 7 • transfer

After a unit, students can repeat the textbook definition of photosynthesis but still say plants get their food from the soil. What does this suggest?

Target word: reverse: diagnose from outcome **Decides:** The prior idea was never confronted.

A	Absolute language	They can, with the right instruction.
B	Correct	Conceptual change needs the prior idea confronted.
C	Inverted operation	Reciting a definition is not understanding.
D	Adjacent category	More memorizing does not change the prior idea.

11. Answer A Concept 8

A teacher is planning a 5E lesson on circuits. ~~The school just bought battery kits, and the class meets after recess.~~ After an Engage hook showing a flashlight that will not light, what should come next?

Target word: 5E sequence **Decides:** Explore comes before Explain.

A	Correct	Explore follows Engage.
B	Confusion set	That is Elaborate, which follows Explain.
C	Adjacent category	Evaluation comes at the end.
D	Inverted operation	Explain comes after exploration in 5E.

12. Answer D Concept 8 • transfer

In a 5E lesson, students use what they learned about circuits to design a doorbell for a model house. Which phase is this?

Target word: reverse: name the phase **Decides:** Applying to a new context.

A	Adjacent category	Builds the formal concept, not its application.
B	Confusion set	Explore comes before formal understanding.
C	Inverted operation	The opening hook.
D	Correct	Extending understanding to new situations.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Erikson's psychosocial stages	School-age children (about 6–12) work on industry vs. inferiority—competence through accomplishment; adolescents work on identity vs. role confusion.	Assigning the identity crisis to elementary students or industry to teens.	An age range plus a social-emotional need.
2	Executive function and the developing brain	The prefrontal cortex, which supports planning, impulse control, and working memory, keeps maturing into the mid-20s, so adolescents benefit from external structure.	Treating poor planning in teens as purely a motivation or character problem.	An adolescent who knows the rule but acts impulsively or misses long-range deadlines.
3	Neuroplasticity, sleep, and stress	The brain changes with practice; sleep consolidates learning, and chronic stress impairs memory and attention.	Believing ability is fixed or that sleep is wasted study time.	A learner trading sleep for study, or a highly stressed learner.
4	Self-efficacy (Bandura)	Self-efficacy grows most from mastery experiences, then from watching similar peers succeed, persuasion, and managing stress.	Relying on general praise to build confidence.	A student who says “I can't do this.”
5	Bronfenbrenner's ecological systems	Microsystem = direct settings (home, class); mesosystem = links between them; exosystem = settings that affect the child indirectly (a parent's job); macrosystem = culture and values.	Calling the family-teacher connection a microsystem.	A relationship or setting to classify.
6	Self-determination theory	Motivation grows when learners feel autonomy, competence, and relatedness.	Relying on rewards alone, which can weaken intrinsic interest.	Designing for lasting motivation.
7	Conceptual change and misconceptions	Students bring prior ideas; teaching must surface them and create evidence that conflicts with them before the new idea can replace them.	Assuming telling the correct answer erases a misconception.	A persistent wrong idea about how the world works.
8	The 5E learning cycle	Engage, Explore, Explain, Elaborate, Evaluate—students explore a phenomenon before formal explanation.	Putting the teacher's explanation before exploration.	Ordering science lesson phases.

Course scope (public description): Learner Development and the Science of Learning provides WGU students with a deep understanding of the science behind learning processes. This course covers a broad spectrum of topics pertaining to the science of learning, including cognitive development, learning theories, neuroscience in education, and the impact of developmental milestones on learning. Students will explore how these concepts apply to learning environments and educational levels, from early childhood through adolescence. The course emphasizes evidence-based practices and the practical application of learning science principles, equipping students with strategies to enhance learning outcomes and learner engagement. This course aims to empower educators to create more effective, inclusive, and engaging learning experiences for all learners. Candidates will evaluate science learning theories in classrooms.